

Building an AI-First Engineering Culture

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Adopting AI in software development is inevitable, but doing so thoughtfully is what separates thriving teams from frustrated ones. This post outlines principles, real-world examples, and practical steps for leaders to integrate AI into developer workflows in ways that enhance trust, creativity, and productivity.

Introduction

Artificial intelligence (AI) is quickly becoming a cornerstone of how development teams build, test, and deliver software. From writing and reviewing code to automating tests and managing deployments, AI promises productivity gains and new opportunities for innovation. Yet amid the excitement, there's always the risk of getting it wrong. Rolling out AI tools too quickly, or without enough thought and experimentation, can leave developers feeling disempowered, frustrated, and distrustful. Left unchecked, AI tools can erode the very culture that makes engineering teams creative and resilient.

An AI-first engineering culture is not simply about adopting tools. It involves rethinking workflows, policies, and mindsets so AI augments developers rather than replacing or overwhelming them. This post outlines practical principles and actions to help engineering leaders, managers, and CTOs infuse AI into their organizations in ways that enhance creativity, trust, and quality.

More Than Tooling Adoption

Engineering has continuously evolved through the introduction of new tools, from version control to cloud platforms. AI is different as it changes the relationship between people and their work in more profound ways. If introduced poorly, AI can lead to:

- Over-reliance, where developers stop questioning suggestions and lose confidence in their judgment.
- Distrust, when AI-generated outputs violate conventions or produce errors with no explanation.
- Compliance and IP problems, as teams unwittingly ship code with unclear licensing or provenance. (Read more on this topic)
- Loss of engagement, when repetitive tasks disappear but meaningful work isn't reimaged.

For these reasons, leaders must look beyond deployment or adoption and consider how to integrate AI in ways that enhance their teams' autonomy and motivation. To do that, you'll need to anchor your approach in a few key principles.

Principles for a Thoughtful AI-First Culture

Every engineering organization is different, but several guiding principles (work in progress) can help ensure that AI complements rather than competes with your developers.

1. **Transparency:** Make AI's role explicit. Developers should understand what data drives the tools, where their suggestions come from, and where the boundaries of reliability lie.
2. **Empowerment:** Position AI as an assistant, not a replacement. Encourage developers to question, override, or improve upon AI outputs rather than blindly accepting them.
3. **Governance:** Establish clear rules about what is and isn't acceptable, especially regarding licensing, security, and code review.
4. **Iteration:** Approach AI adoption as a journey. Start small, gather feedback, and adapt. There's no perfect rollout; it's about continuous learning and improvement.

These principles (consider them a starting point) set the stage for success, but they only matter if you translate them into concrete actions, which we turn to next.

Getting Started

Translating these principles into day-to-day practice starts with a careful assessment of your development workflows and priorities. We will explore five practices:

1. **Audit workflows:** Identify repetitive, low-value tasks that consume developer time. These are good candidates for AI support.

2. **Train your team:** Educate everyone about both the capabilities and limitations of the tools you're introducing.
3. **Preserve rigorous review:** Maintain strong code review and testing processes to catch issues that AI might miss.
4. **Define acceptable use:** Clarify which outputs need human review and when AI-generated code should be rewritten or rejected.
5. **Gather feedback:** Create feedback loops to monitor productivity gains, developer satisfaction, and trust.

Once you've begun implementing these steps, the next challenge is knowing how to measure progress and decide on success metrics.

Measuring Success

It's easy to assume that faster delivery equals success, but a healthy AI-first engineering culture requires a more nuanced view. Organizations typically look for:

- Shorter cycle times for routine tasks, paired with sustained or improved quality.
- Maintained or increased developer satisfaction and engagement.
- Fewer bottlenecks and better focus on high-value work.
- Continued rigor in compliance, security, and governance practices.

Such metrics help ensure that AI is improving both performance and culture, and it's worth learning from companies already on this path.

Real-World Examples

Several companies have already demonstrated how thoughtful integration of AI can enable developers without undermining their creativity or trust:

- **GitHub:** Integrated Copilot to help developers write boilerplate code, tests, and documentation, while keeping human review central and iterating based on developer feedback.
- **Shopify:** Piloted generative AI for documentation, testing, and code scaffolding, positioning it explicitly as a co-pilot rather than a replacement.
- **Microsoft:** Embedded AI in Visual Studio and Azure DevOps pipelines for PR descriptions, duplicate ticket detection, and security checks, all paired with clear training and documentation.
- **Intel:** Leveraged AI-driven static analysis and automated compliance checks to catch vulnerabilities earlier, freeing developers to focus on meaningful work.

- **Salesforce:** Deployed AI agents to triage bug reports and support requests, reducing operational noise for developers.

These few examples demonstrate that success stems from intentional design, not from simply throwing tools into the mix, which leads us to consider what you can do next.

Lead the Culture

AI is no longer optional for engineering organizations, but how you adopt it will define your success far more than when you adopt it.

An AI-first engineering culture is focused on empowering developers. It's about creating an environment where AI reduces toil, reveals opportunities, and strengthens human creativity and judgment.

Companies like GitHub, Shopify, Microsoft, Intel, and Salesforce have demonstrated that thoughtful integration, combined with clear training, effective feedback loops, and robust governance, can deliver measurable productivity gains without compromising trust or quality. As a leader, you can start today by piloting AI in targeted areas, educating your teams, embedding tools seamlessly into workflows, and maintaining a strong focus on both velocity and governance.

- **Start small, measure, and iterate:** Choose a few low-risk, high-impact areas to pilot AI tools. Evaluate impact, then expand gradually.
- **Invest in education:** Help your developers understand *how* to use AI and *when not to*.
- **Embed AI into workflows naturally:** Integrate tools into IDEs, CI/CD, or ticketing systems so developers don't have to context-switch.
- **Balance velocity with governance:** Maintain strong code reviews, clear licensing guidelines, and security checks.
- **Solicit and act on feedback:** Treat the rollout as a dialogue with your team.

At the end of the day, AI is another tool. What truly differentiates organizations is how they lead their people through change and ensure that every developer still feels proud of the work they do.

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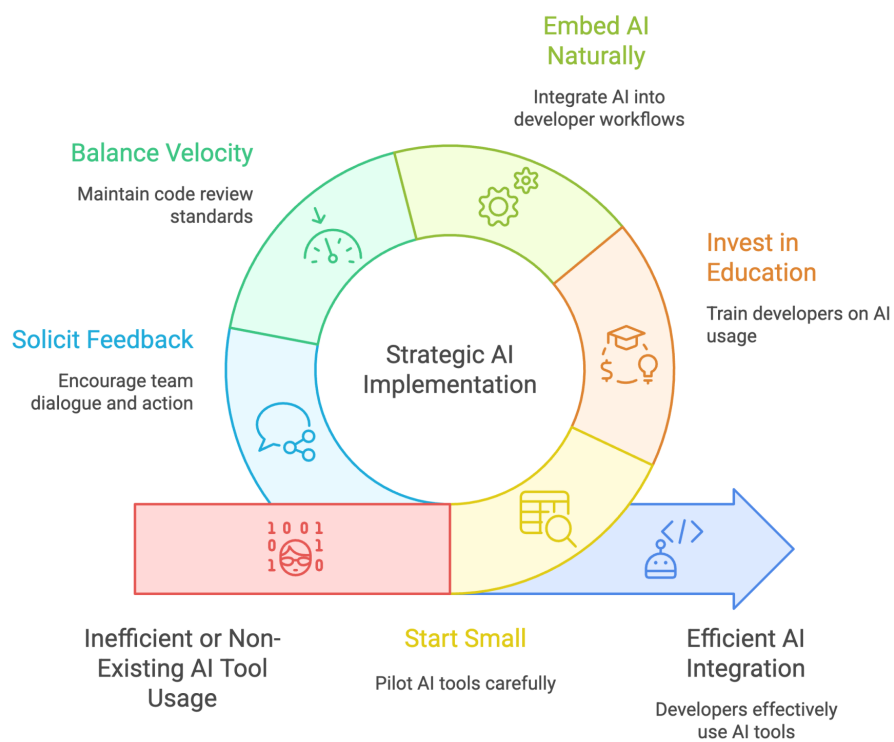


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About the author

Dr. Ibrahim Haddad leads the Infotainment Engineering Group at Volvo Cars. He previously served as Vice President of Strategic Programs (AI) at the Linux Foundation, where he was the Executive Director of LF AI & Data and the Founding Executive Director of the PyTorch Foundation. Previously, Dr. Haddad served as Vice President of R&D and a Distinguished Engineer at Samsung Research. His career spans technical and portfolio leadership roles at Ericsson Research, the Open Source Development Labs, Motorola, Palm, and Hewlett-Packard. ([LinkedIn](#) / [Website](#))